Math 142, S14 **Name:** Answer

Quiz 7 **Section:** **SSN: xxx-xx-**

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Determine whether the sequence $\{a_n\}_{n=1}^{+\infty}$ converges or diverges and find the limit if it converges.

(a) $a_n = \dfrac{2+5n^2}{n+n^2}$

$$\lim_{n\to+\infty} \frac{2+5n^2}{n+n^2} = \lim_{n\to+\infty} \frac{\frac{2+5n^2}{n^2}}{\frac{n+n^2}{n^2}}$$

$$= \lim_{n\to+\infty} \frac{5+\frac{2}{n^2}}{1+\frac{1}{n}}$$

$$= 5$$

$\Rightarrow$ Convergent

(b) $a_n = n^2 e^{-n}$

$$\lim_{n\to+\infty} n^2 e^{-n} = \lim_{x\to+\infty} x^2 e^{-x}$$

$$= \lim_{x\to+\infty} \frac{x^2}{e^x}$$

$$= \lim_{x\to+\infty} \frac{2x}{e^x}$$

$$= \lim_{x\to+\infty} \frac{2}{e^x}$$

$= 0$ $\Rightarrow$ Convergent

Problem 2. (5 points) Determine whether the series is convergent or divergent and find its sum if it is convergent.

(a) $\displaystyle\sum_{n=1}^{+\infty} \frac{(-3)^{n-1}}{4^n}$

$$= \sum_{n=1}^{+\infty} \frac{1}{4}\left(-\frac{3}{4}\right)^{n-1}$$

$$r = -\frac{3}{4} \Rightarrow |r| = \frac{3}{4} < 1$$

$\Rightarrow$ Convergent

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{(-3)^{n-1}}{4^n} = \frac{\frac{1}{4}}{1-\left(-\frac{3}{4}\right)}$$

$$= \frac{\frac{1}{4}}{\frac{7}{4}}$$

$$= \frac{1}{7}$$

(b) $\displaystyle\sum_{n=1}^{+\infty} \frac{1}{2n}$

$$= \frac{1}{2}\underbrace{\sum_{n=1}^{+\infty} \frac{1}{n}}_{}$$ $\rightarrow$ harmonic series

$$= +\infty$$

$\Rightarrow$ divergent